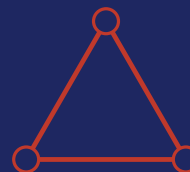
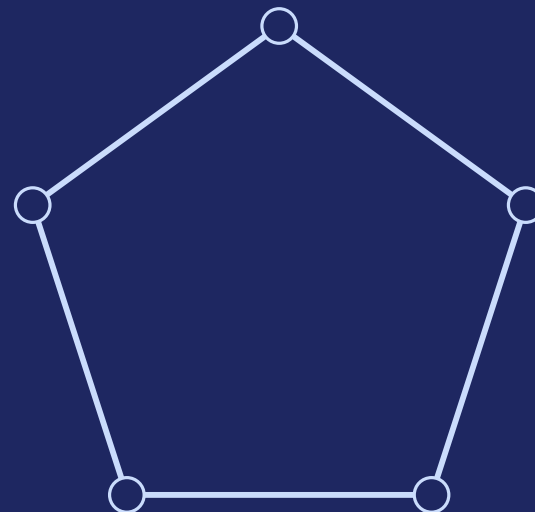


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Extremal combinatorics

Formal proofs in Lean 4



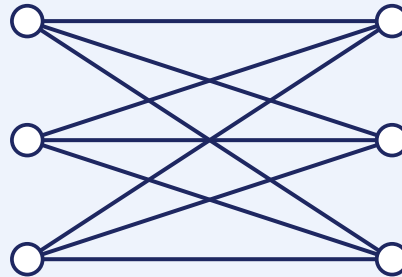
What I work on

Extremal combinatorics

The study of how large or small combinatorial structures can be under given constraints.

Mantel's theorem (1907)

Every triangle-free graph on n vertices has at most $n^2/4$ edges (**edge density $\leq 1/2$**).



extremal: complete bipartite

Flag algebras (by Razborov)

- A mathematical framework for symbolically expressing relationships between combinatorial objects.

Flag-algebra proof of Mantel's theorem

Every triangle-free graph on n vertices has at most $n^2/4$ edges (**edge density $\leq 1/2$**).

$$\begin{aligned} \text{---} &= \frac{1}{3} \overset{\circ}{\text{---}}\overset{\circ}{\text{---}} + \frac{2}{3} \overset{\circ}{\text{---}}\overset{\circ}{\text{---}}\overset{\circ}{\text{---}} \\ &\leq \frac{1}{3} \overset{\circ}{\text{---}}\overset{\circ}{\text{---}} + \frac{2}{3} \overset{\circ}{\text{---}}\overset{\circ}{\text{---}}\overset{\circ}{\text{---}} + \frac{1}{2} \left(\overset{\circ}{\text{---}}\overset{\circ}{\text{---}} - \frac{1}{3} \overset{\circ}{\text{---}}\overset{\circ}{\text{---}} - \frac{1}{3} \overset{\circ}{\text{---}}\overset{\circ}{\text{---}}\overset{\circ}{\text{---}} \right) + \frac{1}{3} \overset{\circ}{\text{---}}\overset{\circ}{\text{---}} \\ &= \frac{1}{2} \cdot \left(\overset{\circ}{\text{---}}\overset{\circ}{\text{---}} + \overset{\circ}{\text{---}}\overset{\circ}{\text{---}} + \overset{\circ}{\text{---}}\overset{\circ}{\text{---}}\overset{\circ}{\text{---}} \right) = \frac{1}{2} \cdot \mathbf{1}. \end{aligned}$$

Flag algebras (by Razborov)

- Reduces optimization problems in extremal combinatorics to computer-solvable **semidefinite programming (SDP)** instances.

Flag algebras (by Razborov)

- Reduces optimization problems in extremal combinatorics to computer-solvable **semidefinite programming (SDP)** instances.
- Has resolved a wide range of open problems.
 - Triangle-density problem
 - Erdős pentagon problem
 - Density of induced 5-cycle
 - Exact values of some Ramsey numbers
 - Turán's tetrahedron problem
 - Rainbow-triangle density of 3-colored graphs
 - Quasirandom-forcing permutation-set problem

Flag algebras (by Razborov)

- **Flagmatic (by Vaughan)**

Software that automates SDP formulation and solving, enabling the discovery of new upper bounds on extremal densities.

```
jgyw0910@BOOK-B4UB0DPUD4:~/flagmatic-1.5.1$ ./flagmatic --r 2 --n 5 --forbid 3.3 --dir output
flagmatic version 1.5
=====
Forbidding 3.3
Using admissible graphs of order 5.
Generated 1 type of order 1, with 5 flags of order 3.
Generated 3 types of order 3, with [8, 6, 5] flags of order 4.
Generated 14 admissible graphs.
Approximate floating-point bound is 0.50000000
```

Reproducing Mantel's theorem in Flagmatic

Lean 4 formalization

with Seonghun Park, Jihoon Hyun, Sang-il Oum, Hongseok Yang

Specification layer

Flag algebra theory formalization

Jan 2025 – Oct 2025

15,775

lines

Reflection layer

Computable graph enumerations + density computations

Oct 2025 – Feb 2026

3,895

lines

Flagmatic-to-Lean framework

Flagmatic output → formal Lean proof, semi-automatic

Mar 2026 – Jun 2026

9,868

lines

Meta-theory of constraints

Razborov builds constraints (e.g. triangle-free) into the algebra;
we add them afterwards — and prove when the two agree.

Jun 2026

14,763

lines

= 44,301 lines of Lean 4

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Codex & Claude Code wrote almost all of it

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fully autoformalized by Claude Code

= 44,301 lines of Lean 4

Verified results

Result	Forbidden	Verified bound
Mantel's theorem	K_3	edge density $\leq 1/2$
Turán bound for K_4	K_4	edge density $\leq 2/3$
Turán bound for K_5	K_5	edge density $\leq 3/4$
P_3 density in triangle-free graphs	K_3	P_3 density $\leq 3/4$
C_4 density in triangle-free graphs	K_3	C_4 density $\leq 3/8$
Erdős pentagon theorem	K_3	C_5 density $\leq 24/625$
Edge density in C_5 -free graphs	C_5	edge density $\leq 1/2$

What I want to try next

- Autoformalize flag-algebra proofs beyond Flagmatic.
- Develop AI-guided proof methods to overcome the scalability limits of flag-algebra proofs.
- Pair the framework with an AI model that finds extremal constructions, aiming to determine exact bounds automatically.